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### Insertion apparatus for use with rotary machines

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#### Abstract

An insertion apparatus for use with a rotary machine includes a body extending from an insertion end to a steering end and sized to fit within an annular cavity. The body has a first stiffness and curves along a circumference of the annular cavity as the insertion end travels through the annular cavity. The insertion apparatus also includes a stiffener coupled to the body and extending from the steering end to the insertion end. The stiffener has a second stiffness greater than the first stiffness. The insertion apparatus further includes at least one maintenance device coupled to the insertion end of the insertion apparatus and a displacement mechanism configured to adjust a position of the at least one maintenance device relative to the body.

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## **Background/Summary**

### **CROSS REFERENCE TO RELATED APPLICATIONS**

(1) This application claims priority to Indian Patent Application Serial No. 201941040204 filed Oct. 4, 2019, entitled INSERTION APPARATUS FOR USE WITH ROTARY MACHINES, which is hereby incorporated by reference in its entirety.

### **BACKGROUND**

(2) The field of the disclosure relates generally to an insertion apparatus and, more particularly, to an insertion apparatus for use with rotary machines having annular cavities.

(3) At least some known rotary machines, such as turbines for aircraft engines and gas and steam powered turbines for industrial applications, include an outer case and at least one rotor that carries multiple stages of rotating airfoils, i.e., blades, which rotate with respect to the outer case. In addition, the outer case carries multiple stages of stationary airfoils, i.e., guide vanes. The blades and guide vanes are arranged in alternating stages. In at least some known rotary machines, shrouds are disposed on the radially inner surfaces of a stator to form a ring seal around tips of the blades.

(4) At least some rotary machines are inspected periodically to determine if components of the rotary machines need repair and/or replacement. However, some components of the rotary machine may be difficult to access and inspect without disassembly of the rotary machine. For example, some components, such as clips attaching the shrouds to shroud hangers, are positioned in annular cavities extending circumferentially around the shrouds. However, the size, shape, and location of the annular cavities prevent at least some known inspection apparatus from accessing and inspecting the clips. Moreover, the annular cavities may be obstructed by components, such as the clips, that extend at least partially into the annular cavities.

(5) Accordingly, it is desirable to provide an insertion apparatus for use with a rotary machine that is configured to inspect components positioned along an annular cavity of the rotary machine.

#### BRIEF DESCRIPTION

(6) In one aspect, an insertion apparatus for use with a rotary machine is provided. The rotary machine defines an annular cavity extending along a circumference. The insertion apparatus includes an insertion end positionable within the annular cavity and configured to travel through the annular cavity. The insertion apparatus also includes a steering end opposite the insertion end. The insertion apparatus further includes a body extending from the insertion end to the steering end and sized to fit within the annular cavity. The body has a first stiffness and curves along the circumference as the insertion end travels through the annular cavity. The insertion apparatus also includes a stiffener coupled to the body and extending from the steering end to the insertion end. The stiffener has a second stiffness greater than the first stiffness. The insertion apparatus further includes at least one maintenance device coupled to the insertion end of the insertion apparatus and a displacement mechanism configured to adjust a position of the at least one maintenance device relative to the body.

(7) In another aspect, a system for use with a rotary machine is provided. The rotary machine defines an annular cavity extending along a circumference. The system includes an insertion apparatus including a body extending from an insertion end to a steering end. The body is sized to fit within the annular cavity and is curved along the circumference. The system also includes at least one maintenance device coupled to the insertion end of the insertion apparatus and a displacement mechanism configured to adjust a position of the at least one maintenance device relative to the body. The system further includes a guide tube positionable in a port of the rotary machine and configured to define a path for the insertion apparatus into the annular cavity. The guide tube is curved to form an at least partially helical path around objects and is sized to extend adjacent an opening into the annular cavity.

(8) In yet another aspect, a method of inspecting a rotary machine is provided. The rotary machine defines an annular cavity extending along a circumference. The method includes positioning a guide tube in a port of the rotary machine to define a path for an insertion apparatus into the annular cavity. The guide tube is curved to form an at least partially helical shape. The method also includes positioning the insertion apparatus along the path and into the annular cavity. The insertion apparatus includes a body having an insertion end and a steering end opposite the insertion end. The body is sized to fit within the annular cavity and is curved along the circumference. The method further includes directing the insertion end of the insertion apparatus through the annular cavity. At least one maintenance device is coupled to the insertion end of the insertion apparatus. The method also includes positioning the at least one maintenance device adjacent a portion of the rotary machine using a displacement mechanism configured to adjust a position of the at least one maintenance device relative to the body.

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## Description

#### DRAWINGS

(1) These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

(2) FIG. 1 is a cross-sectional schematic view of an exemplary rotary machine;

(3) FIG. 2 is a schematic view of a system including an insertion apparatus and a guide tube positioned within a primary cavity of the rotary machine shown in FIG. 1;

(4) FIG. 3 is a schematic view of the insertion apparatus shown in FIG. 2 traveling along the annular cavity of the rotary machine shown in FIG. 1;



- (5) FIG. 4 is a side view of a portion of the insertion apparatus shown in FIGS. 2 and 3;
  - (6) FIG. 5 is a perspective view of the guide tube shown in FIG. 2;
  - (7) FIG. 6 is a perspective view of an alternative embodiment of a displacement mechanism for use with the insertion apparatus shown in FIGS. 2-4; and
  - (8) FIG. 7 is a flow chart of an exemplary method of inspecting a rotary machine.
- (9) Unless otherwise indicated, the drawings provided herein are meant to illustrate features of embodiments of this disclosure. These features are believed to be applicable in a wide variety of systems comprising one or more embodiments of this disclosure. As such, the drawings are not meant to include all conventional features known by those of ordinary skill in the art to be required for the practice of the embodiments disclosed herein.

#### DETAILED DESCRIPTION

- (10) In the following specification and the claims, reference will be made to a number of terms, which shall be defined to have the following meanings.
- (11) The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.
- (12) “Optional” or “optionally” means that the subsequently described event or circumstance may or may not occur, and that the description includes instances where the event occurs and instances where it does not.
- (13) Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about”, “approximately”, and “substantially”, are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Here and throughout the specification and claims, range limitations may be combined and/or interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise.
- (14) Embodiments described herein provide a system for use with a rotary machine having an annular cavity. The system includes an insertion apparatus configured to position a maintenance device along the annular cavity of the rotary machine. In addition, in some embodiments, the system includes a guide tube positionable in a port of the rotary machine and configured to guide the insertion apparatus into the annular cavity. For example, the guide tube is curved and has a helical shape. During deployment, the insertion apparatus is inserted along a path defined by the guide tube into the annular cavity and directed along the annular cavity. The insertion apparatus is sized and shaped to fit within and move along the annular cavity. The maintenance device is coupled to an insertion end of the insertion apparatus and is positionable relative to the rotary machine using a displacement mechanism. As a result, the system allows inspection and/or repair of components at locations within the annular cavity of the rotary machine.
- (15) FIG. 1 is a cross-sectional schematic view of an exemplary rotary machine. In the exemplary embodiment, the rotary machine includes a turbine assembly **100**. In alternative embodiments, the rotary machine includes any assembly. For example, in some embodiments, the rotary machine includes, without limitation, any of the following: a compressor, a blower, a pump, a turbine, a motor, and a generator.
- (16) In the exemplary embodiment, turbine assembly **100** includes an inlet **102**, a compressor **104**, a combustor **106**, a turbine **108**, an outer case **110**, and an exhaust **112**. Fluid flows from inlet **102**, through compressor **104**, through combustor **106**, through turbine **108** and is discharged through exhaust **112**. Also, in the exemplary embodiment, compressor **104** and turbine **108** include airfoils configured to direct fluid through turbine assembly **100**. In particular, compressor **104** and turbine **108** include blades **114**, **116** and guide vanes **118**, **120**. Together, blades **114**, **116**, guide vanes **118**, **120**, and shrouds **122** (shown in FIG. 2) define a primary flowpath **124** of turbine assembly **100**. This flowpath, combined with a flowpath through combustor **106**, defines a primary cavity within

turbine assembly **100**. In alternative embodiments, turbine assembly **100** is configured in any manner that enables turbine assembly **100** to operate as described herein.

(17) Blades **114**, **116** are operably coupled with rotating shafts **126**, **128** such that blades **114**, **116** rotate when rotating shafts **126**, **128** rotate. Accordingly, blades **114**, **116** and rotating shafts **126**, **128** form a rotor of turbine assembly **100**. Guide vanes **118**, **120** and shrouds **122** are stationary components and are coupled to an interior surface **130** of outer case **110**. Blades **114**, **116** and guide vanes **118**, **120** are generally positioned alternately along the rotor axis within turbine assembly **100**. In alternative embodiments, compressor **104** and/or turbine **108** includes any airfoils that enable turbine assembly **100** to operate as described herein.

(18) FIG. 2 is a schematic view of a system **132** including an insertion apparatus **134** and a guide tube **136** positioned in the primary cavity of turbine assembly **100**. FIG. 3 is a schematic view of insertion apparatus **134** traveling along an annular cavity **138** of turbine assembly **100**. Annular cavity **138** extends along a circumference of a respective shroud **122** of turbine assembly **100**.

(19) In the exemplary embodiment, shrouds **122** are at least partially supported by a hanger and extend circumferentially around blades **114**, **116** and alongside a turbine central frame **140**. Clips **142** couple shrouds **122** to hangers and secure shrouds **122** in position during operation of turbine assembly **100**. In the exemplary embodiment, clips **142** are C-shaped and extend at least partially into annular cavity **138**. Clips **142** are spaced apart along the circumference of each shroud **122**. Accordingly, clips **142** are difficult to access from the exterior of turbine assembly **100** or from the primary cavity of turbine assembly **100**. Insertion apparatus **134** is sized and shaped to fit within annular cavity **138** and facilitates inspection and/or repair at locations within annular cavity **138** that are difficult to access from an exterior of turbine assembly **100** by conventional means, such as using a borescope tool. Specifically, in contrast to at least some known insertion apparatus, insertion apparatus **134** is configured to extend into annular cavity **138** to facilitate inspection of clips **142**.

(20) During operation, insertion apparatus **134** enters turbine assembly **100** through any suitable access port or opening of turbine assembly **100**. For example, in some embodiments, insertion apparatus **134** enters and/or exits turbine assembly **100** through any of inlet **102** (shown in FIG. 1), exhaust **112** (shown in FIG. 1), and/or an access port, such as an igniter, borescope, or fuel nozzle port. In the exemplary embodiment, insertion apparatus **134** is sized and shaped to fit within turbine assembly **100** and to travel along annular cavity **138**. For example, insertion apparatus **134** has a height, length, and width that are less than a clearance required to fit within annular cavity **138**. In alternative embodiments, insertion apparatus **134** is any size and shape that enables insertion apparatus **134** to operate as described herein.

(21) Also, in the exemplary embodiment, guide tube **136** extends through a port of turbine assembly **100** and defines a path for insertion apparatus **134**. For example, guide tube **136** defines an interior space sized to receive insertion apparatus **134** and allow insertion apparatus **134** to travel through guide tube **136**. Guide tube **136** may be fixed to turbine assembly **100** by a flange (not shown) coupled to a port of turbine assembly **100**. For example, in some embodiments, a flange extends around guide tube **136** and is sized to fit onto a port of turbine assembly **100**. In alternative embodiments, guide tube **136** is coupled to turbine assembly **100** in any manner that enables guide tube **136** to operate as described herein.

(22) In addition, in the exemplary embodiment, guide tube **136** is configured to direct insertion apparatus **134** into annular cavity **138** of turbine assembly **100**. For example, guide tube **136** is curved and defines a curved path for insertion apparatus **134**. Specifically, guide tube **136** is curved in three-dimensions and has a helical shape. Accordingly, guide tube **136** is able to guide insertion apparatus **134** around obstacles in the primary cavity of turbine assembly **100**. In addition, guide tube **136** is sized such that a tip **144** of guide tube **136** is positioned proximate a target area within turbine assembly **100**, e.g., an opening into annular cavity **138**. In alternative embodiments, guide tube **136** is any size and shape that enables guide tube **136** to operate as described herein.

(23) During operation, insertion apparatus **134** is used to inspect and/or repair any interior components of turbine assembly **100**. For example, in some embodiments, insertion apparatus **134** is positioned adjacent a portion of interior surface **130** of turbine assembly **100** within annular cavity **138**. Interior surface **130** may include a surface of clips **142**. In some embodiments, insertion apparatus **134** detects a characteristic of interior surface **130**. For example, in some embodiments, insertion apparatus **134** is used to generate an image of interior surface **130** and the image is examined to determine the condition of turbine assembly **100** and assess whether repairs are necessary. In further embodiments, insertion apparatus **134** includes a sensor that detects characteristics of interior surface **130**. If repairs are necessary, in some embodiments, insertion apparatus **134** is used to repair interior surface **130**. After inspection and/or repair of interior surface **130**, insertion apparatus **134** exits turbine assembly **100** through any suitable access port or opening of turbine assembly **100**, such as via the route of entry.

(24) FIG. 4 is a side view of a portion of insertion apparatus **134**. Insertion apparatus **134** includes an insertion end **146** and a steering end **148** opposite insertion end **146**. Insertion end **146** is positionable within annular cavity **138** (shown in FIG. 3) of turbine assembly **100** (shown in FIG. 1) at target locations, such as adjacent clips **142** (shown in FIG. 3). In some embodiments, insertion end **146** is shaped to facilitate movement of insertion apparatus **134** through annular cavity **138** (shown in FIG. 3) without insertion apparatus **134** being caught on objects. For example, in some embodiments, insertion end **146** is curved. In further embodiments, insertion end **146** is angled, e.g., insertion end **146** has a V-shape.

(25) Also, in the exemplary embodiment, insertion apparatus **134** includes at least one maintenance device **150** coupled to insertion end **146** of insertion apparatus **134** to allow insertion apparatus **134** to perform an inspection and/or repair operation within annular cavity **138** (shown in FIG. 3) of turbine assembly **100** (shown in FIG. 1). In some embodiments, maintenance device **150** includes at least one sensor **152** that is configured to contact surfaces. For example, in some embodiments, sensor **152** is an eddy current sensor. In alternative embodiments, insertion apparatus **134** includes any maintenance device **150** that enables insertion apparatus **134** to operate as described herein. For example, in some embodiments, maintenance device **150** of insertion apparatus **134** includes, without limitation, any of the following: an applicator, a drill, a grinder, a heater, a welding electrode, a sprayer, an optical sensor (e.g., visible, infrared, and/or multi-spectral sensor), a mechanical sensor (e.g., stylus profilometer, coordinate measurement probe, load transducer, linear variable differential transformer), a thermal sensor (e.g., pyrometer, thermocouple, resistance temperature detector), a magnetic sensor, an acoustic sensor (e.g., piezoelectric, microphone, ultrasound), and an electromagnetic sensor (e.g., eddy current, potential drop, x-ray).

(26) In addition, in the exemplary embodiment, maintenance device **150** is positionable in a plurality of orientations using a displacement mechanism **154**. For example, in the exemplary embodiment, maintenance device **150** is pivotably coupled to insertion end **146** of insertion apparatus **134**. Displacement mechanism **154** includes a cable **156** extending from maintenance device **150** to steering end **148**. Maintenance device **150** is selectively pivoted by manipulating cable **156**. For example, maintenance device **150** is positionable between a first orientation in which maintenance device **150** is aligned with a translation direction **158** of insertion apparatus **134** and a second orientation in which insertion apparatus **134** extends at an angle relative to translation direction **158**. The ability to position maintenance device **150** in a plurality of orientations facilitates precise positioning of maintenance device **150** relative to a target location. In alternative embodiments, maintenance device **150** is positionable in any manner that enables insertion apparatus **134** to operate as described herein.

(27) Also, in the exemplary embodiment, insertion apparatus **134** includes a body **160** extending from insertion end **146** to steering end **148**. Body **160** is sized and shaped to fit within annular cavity **138** (shown in FIG. 3). For example, body **160** includes a first surface **162**, a second surface **164** opposite first surface **162**, and edges **166** extending along first surface **162** and second surface

**164.** First surface **162** and second surface **164** extend from steering end **148** to insertion end **146** and define a thickness **168** of body **160** therebetween. Thickness **168** is less than a width of each of first surface **162** and second surface **164**. For example, in some embodiment, thickness **168** of body **160** is less than about 0.02 inches. In addition, first surface **162** and second surface **164** are substantially smooth, e.g., free of bumps, perforations, folds, or other surface features. Accordingly, body **160** is ribbon-shaped. In alternative embodiments, body **160** has any shape that enables insertion apparatus **134** to function as described herein.

(28) Moreover, in the exemplary embodiment, body **160** is flexible and conforms to a curve of annular cavity **138**. Specifically, body **160** has a first stiffness and is configured to curve in a direction perpendicular to first surface **162** and second surface **164** along the curve of annular cavity **138**. Accordingly, insertion apparatus **134** is able to travel within annular cavity **138** circumferentially along shroud **122** (shown in FIG. 2). Also, body **160** has a length that is greater than a circumference of shroud **122** such that insertion apparatus **134** is configured to extend from the exterior of turbine assembly **100** (shown in FIG. 2), into annular cavity **138** (shown in FIG. 2) and through the entirety of annular cavity **138**. In the exemplary embodiment, body **160** includes at least one conductor **170**, e.g., a flexible metal wire, encased within insulation **172**. In alternative embodiments, insertion apparatus **134** includes any body **160** that enables insertion apparatus **134** to operate as described herein.

(29) In addition, in the exemplary embodiment, insertion apparatus **134** includes a stiffener **174** coupled to and extending along body **160** from steering end **148** to insertion end **146**. In some embodiments, stiffener **174** is coupled to body **160** or formed integrally with body **160**. Stiffener **174** is configured to resist bending of body **160** and facilitate translation and steering of insertion apparatus **134** within annular cavity **138** (shown in FIG. 3). Moreover, stiffener **174** prevents body **160** from folding as insertion apparatus **134** is moved. Stiffener **174** has a second stiffness greater than the first stiffness of body **160**. For example, in some embodiments, stiffener **174** includes a material that has a greater stiffness per unit volume than the material of body **160**. In further embodiments, stiffener **174** has a thickness greater than thickness **168** of body **160**. In the exemplary embodiment, stiffener **174** includes at least one cable and/or at least one hollow tube extending from steering end **148** to insertion end **146**. Stiffener **174** has a width that is less than a width of body **160** and allows some flexing of body **160**. In alternative embodiments, insertion apparatus **134** includes any stiffener **174** that enables insertion apparatus **134** to operate as described herein. In further embodiments, stiffener **174** is omitted.

(30) In some embodiments, body **160** and/or stiffener **174** is configured to convey power and communication signals for maintenance device **150**. In further embodiments, maintenance device **150** transmits and receives signals and/or power using any wired and/or wireless connections. For example, in some embodiments, a component, such as a harness or tether, extends from maintenance device **150** to the exterior of turbine assembly **100** and provides power to maintenance device **150**, allows maintenance device **150** to send and/or receive signals, and/or transmits mechanical force, fluids, or thermal energy to maintenance device **150**.

(31) FIG. 5 is a perspective view of guide tube **136**. Guide tube **136** includes a body **176** and tip **144**. Together body **176** and tip **144** define an interior cavity **178** sized to receive insertion apparatus **134** (shown in FIG. 4). Guide tube **136** is configured to guide insertion apparatus **134** (shown in FIG. 2) toward annular cavity **138** (shown in FIG. 2). Body **176** is substantially rigid relative to insertion apparatus **134** (shown in FIG. 2) such that body **176** does not deform when insertion apparatus **134** travels through guide tube **136**. In addition, body **176** is curved in 3-dimensions and forms a helical shape. When positioned in a port of turbine assembly **100** (shown in FIG. 2), guide tube **136** defines a path for insertion apparatus **134** around components of turbine assembly **100**, such as blades **114** (shown in FIG. 2), and into annular cavity **138** (shown in FIG. 2). In alternative embodiments, guide tube **136** includes any body **176** that enables guide tube **136** to function as described herein. For example, in some embodiments, body **176** is segmented and the

segments of body **176** are positionable relative to each other.

(32) Also, in the exemplary embodiment, tip **144** is pliable and has a stiffness less than the stiffness of body **176**. In addition, tip (**144**) is relatively flexible compared to the rest of guide tube **136**. Accordingly, tip **144** may deform when contacted by insertion apparatus **134**. Tip **144** facilitates guiding insertion apparatus **134** into the opening of annular cavity **138** when guide tube **136** is positioned proximate annular cavity **138**. In alternative embodiments, guide tube **136** includes any tip that enables guide tube **136** to function as described herein.

(33) FIG. **6** is a perspective view of an alternative embodiment of a displacement mechanism **200** for use with the insertion apparatus **134** (shown in FIGS. **2-4**). Displacement mechanism **200** is configured to adjust the orientation of maintenance device **150** (shown in FIG. **4**) relative to insertion end **146** (shown in FIG. **4**) of insertion apparatus **134**. For example, displacement mechanism **200** includes an inflatable bladder **202** and a flexible tube **204** configured to deliver a fluid to inflatable bladder **202**. Inflatable bladder **202** is configured to couple to maintenance device **150** (shown in FIG. **4**). In addition, inflatable bladder **202** is configured to switch between a deflated position and an at least partially inflated position to adjust the position of maintenance device **150** relative to body **160** (shown in FIG. **4**) of insertion apparatus **134** (shown in FIG. **4**).

(34) Also, in the exemplary embodiment, flexible tube **204** extends from insertion end **146** (shown in FIG. **4**) to steering end **148** (shown in FIG. **4**) and is configured to carry fluid such as air, water, and/or any other suitable fluid to inflatable bladder **202**. In some embodiments, displacement mechanism **200** includes a valve or other regulatory mechanism to allow control of the fluid flow through flexible tube **204**. In some embodiments, flexible tube **204** provides stiffness to body **160** (shown in FIG. **4**) and acts as stiffener **174** (shown in FIG. **4**) of insertion apparatus **134** (shown in FIG. **4**).

(35) FIG. **7** is a flow chart of an exemplary method **300** of inspecting turbine assembly **100** (shown in FIG. **1**). In reference to FIGS. **1-3** and **7**, method **300** includes positioning **302** guide tube **136** in a port of the rotary machine to define a path for insertion apparatus **134** into annular cavity **138** defined by shroud **122**. For example, in some embodiments, a flange fits onto a port of turbine assembly **100** and couples guide tube **136** to turbine assembly **100**. Guide tube **136** defines a path from an exterior of turbine assembly **100**, through the port, and into annular cavity **138**. Guide tube **136** is shaped to curve around obstacles within turbine assembly **100** and is sized such that tip **144** of guide tube **136** is positioned adjacent an opening into annular cavity **138**. In alternative embodiments, guide tube **136** defines any path that enables system **132** to operate as described herein.

(36) In addition, method **300** includes positioning **304** insertion apparatus **134** along the path and into annular cavity **138**. For example, in some embodiments, insertion apparatus **134** is inserted into the interior cavity **178** of guide tube **136** and moved through guide tube **136**. Guide tube **136** guides insertion apparatus **134** around obstacles within turbine assembly **100** and into annular cavity **138** as insertion apparatus **134** is moved along the path defined by guide tube **136**. In some embodiments, method **300** includes deforming tip **144** to guide insertion apparatus **134** into annular cavity **138**. For example, in some embodiments, tip **144** has a stiffness less than the stiffness of body **176** and allows insertion apparatus **134** to transition smoothly from guide tube **136** and into annular cavity **138**.

(37) Also, method **300** includes directing **306** insertion end **146** of insertion apparatus **134** through annular cavity **138**. For example, in some embodiments, insertion end **146** of insertion apparatus **134** is directed through annular cavity **138** using a steering interface located at steering end **148**. In some embodiments, insertion end **146** is directed by moving body **160** along translation direction **158**. Insertion end **146** moves along the circumference of shroud **122** within annular cavity **138** as body **160** is moved in translation direction **158**. After insertion end **146** has traveled around substantially the entire circumference of shroud **122** or has passed a desired target location, insertion apparatus **134** may be moved backwards along translation direction **158** within annular

cavity **138**. In alternative embodiments, insertion apparatus **134** is moved in any manner that enables insertion apparatus **134** to operate as described herein.

(38) Also, method **300** includes positioning **308** maintenance device **150** coupled to insertion end **146** of insertion apparatus **134** adjacent a portion of the rotary machine using displacement mechanism **154**. For example, in some embodiments, cable **156** of displacement mechanism **154** is used to pivot maintenance device **150** about insertion end **146** of insertion apparatus **134** such that maintenance device **150** contacts interior surface **130** of clips **142**. In further embodiments, inflatable bladder **202** (shown in FIG. **6**) is inflated/deflated to adjust the position of maintenance device **150**.

(39) Moreover, method **300** includes detecting **310** a characteristic of an interior portion of the rotary machine using maintenance device **150**. For example, in some embodiments, maintenance device **150** includes sensor **152** with electrodes that contact clips **142** within annular cavity **138** to determine a characteristic of clips **142**. Displacement mechanism **154** facilitates maintenance device **150** properly contacting clips **142** or other components of turbine assembly **100**.

Accordingly, insertion apparatus **134** is able to provide accurate and reliable information for components along annular cavity **138**.

(40) The above described embodiments provide a system for use with a rotary machine having an annular cavity. The system includes an insertion apparatus configured to position a maintenance device along the annular cavity of the rotary machine. In addition, in some embodiments, the system includes a guide tube positionable in a port of the rotary machine and configured to guide the insertion apparatus into the annular cavity. For example, the guide tube is curved and has a helical shape. During deployment, the insertion apparatus is inserted along a path defined by the guide tube into the annular cavity and directed along the annular cavity. The insertion apparatus is sized and shaped to fit within and move along the annular cavity. The maintenance device is coupled to an insertion end of the insertion apparatus and is positionable relative to the rotary machine using a displacement mechanism. As a result, the system allows inspection and/or repair of components at locations within the annular cavity of the rotary machine.

(41) An exemplary technical effect of the methods, systems, and apparatus described herein includes at least one of: (a) reducing the time to inspect and/or repair a rotary device or other applicable circular mechanism; (b) increasing the accessibility of difficult-to-reach locations within a turbine assembly for inspection and/or in situ repair; (c) reducing the time that circular mechanisms are out of service for maintenance; (d) increasing the precision and/or reliability of inspection and repair of circular mechanisms; (e) reducing unplanned service outages for a circular mechanisms; (f) enhancing data capture for use in quantifying and/or modeling the service condition of at least some components of the circular mechanism; and (g) providing a system and method for inspecting shroud hanger clips within an annular cavity.

(42) Exemplary embodiments of methods and systems for use with rotary machines are not limited to the specific embodiments described herein, but rather, components of systems and/or steps of the methods may be utilized independently and separately from other components and/or steps described herein. For example, the methods and systems may also be used in combination with other systems requiring inspection and/or repair of components, and are not limited to practice with only the systems and methods as described herein. Rather, the exemplary embodiment can be implemented and utilized in connection with many other applications, equipment, and systems that may benefit from using a service apparatus for inspection and/or repair.

(43) Although specific features of various embodiments of the disclosure may be shown in some drawings and not in others, this is for convenience only. In accordance with the principles of the disclosure, any feature of a drawing may be referenced and/or claimed in combination with any feature of any other drawing.

(44) This written description uses examples to disclose the embodiments, including the best mode, and also to enable any person skilled in the art to practice the embodiments, including making and

using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

## Claims

1. An insertion apparatus for use with a rotary machine, the rotary machine defining an annular cavity extending along a circumference, said insertion apparatus comprising: an insertion end positionable within the annular cavity and configured to travel through the annular cavity; a steering end opposite said insertion end; a body extending from said insertion end to said steering end and sized to fit within the annular cavity, wherein said body includes a first material having a first material stiffness and curves along the circumference as said insertion end travels through the annular cavity; a stiffener integrally formed and fixed with respect to said body and extending from said steering end to said insertion end, wherein said stiffener includes a second material having a second material stiffness that is greater than the first material stiffness and is configured to resist bending of said body; wherein the stiffener is fixed adjacent to one edge of the body; wherein the edge of the body corresponding to the location of the stiffener leads an opposite edge of the body that does not correspond to the stiffener during insertion into the annular cavity; wherein the stiffener is configured to resist bending and prevent the body from folding when being translated into the annular cavity; wherein the body and stiffener comprise a uniform and uninterrupted cross-section extending from the insertion end to the steering end; wherein (1) arrangement of the stiffener fixed and adjacent to one edge of the body and (2) integral formation of the body with the stiffener are cooperatively effective to provide the body with an overall stiffness, the overall stiffness being constant and unchanged prior to deformation of the body and after deformation of the body as the body curves along the circumference during insertion and movement of the body through the annular cavity; wherein the stiffener is uncontrolled during insertion and movement of the body through the annular cavity; at least one maintenance device coupled to said insertion end of said insertion apparatus; a displacement mechanism configured to adjust a position of said at least one maintenance device relative to said body; a guide tube having a segmented body including segments that are positionable relative to each other, the body and stiffener axially movable through the segments of the guide tube; and wherein the guide tube is curved to form an at least partially helical shape.
2. The insertion apparatus in accordance with claim 1, wherein said body comprises a first surface and a second surface opposite said first surface, said first surface and said second surface extending from said steering end to said insertion end, said body having a thickness defined between said first surface and said second surface, wherein the thickness is less than a width of said first surface and said second surface, and wherein said body is configured to curve in a direction perpendicular to said first surface and said second surface.
3. The insertion apparatus in accordance with claim 1, wherein said at least one maintenance device is pivotably coupled to said insertion end, and wherein said displacement mechanism is configured to pivot said at least one maintenance device about said insertion end.
4. The insertion apparatus in accordance with claim 1, wherein said body is configured to convey power and communication signals for said at least one maintenance device.
5. The insertion apparatus in accordance with claim 1, wherein said at least one maintenance device comprises at least one of the following: an optical sensor, a mechanical sensor, a thermal sensor, a magnetic sensor, an acoustic sensor, and an electromagnetic sensor.
6. The insertion apparatus in accordance with claim 1, wherein the stiffener has a stiffener thickness greater than a thickness of the body.

7. The insertion apparatus in accordance with claim 1, wherein the body is a ribbon-shaped body.
8. The insertion apparatus in accordance with claim 7, wherein the body has a thickness less than about 0.02 inches.
9. The insertion apparatus in accordance with claim 1, wherein the guide tube has a substantially rigid body and a pliable tip at a guide tube insertion end.
10. The insertion apparatus in accordance with claim 1, wherein the body and the guide tube both comprise a substantially rectangular cross-section.
11. An insertion apparatus for use with a rotary machine, the rotary machine defining an annular cavity extending along a circumference, said insertion apparatus comprising: an insertion end positionable within the annular cavity and configured to travel through the annular cavity; a steering end opposite said insertion end; a body extending from said insertion end to said steering end and sized to fit within the annular cavity, wherein said body has a first stiffness, wherein said body comprises a first continuous surface and a second continuous surface opposite the first continuous surface that curve along the circumference as said insertion end travels through the annular cavity; a stiffener integrally formed and fixed with respect to said body and extending from said steering end to said insertion end, wherein said stiffener has a second stiffness greater than the first stiffness and is configured to resist bending of said body; wherein the stiffener is fixed adjacent to one edge of the body; wherein the edge of the body corresponding to the location of the stiffener leads an opposite edge of the body that does not correspond to the stiffener during insertion into the annular cavity; wherein the stiffener is configured to resist bending and prevent the body from folding when being translated into the annular cavity; wherein the body and stiffener comprise a uniform and uninterrupted cross-section extending from the insertion end to the steering end; wherein (1) arrangement of the stiffener fixed and adjacent to one edge of the body and (2) integral formation of the body with the stiffener are cooperatively effective to provide the body with an overall stiffness, the overall stiffness being constant and unchanged prior to deformation of the body and after deformation of the body as the body curves along the circumference during insertion and movement of the body through the annular cavity; wherein the stiffener is uncontrolled during insertion and movement of the body through the annular cavity; at least one maintenance device coupled to said insertion end of said insertion apparatus; a displacement mechanism configured to adjust a position of said at least one maintenance device relative to said body; and a guide tube having a segmented body including segments that are positionable relative to each other, the body and stiffener axially movable through the segments of the guide tube, wherein the guide tube is curved to form an at least partially helical shape; and wherein said first continuous surface and said second continuous surface extend from said steering end to said insertion end, said body having a thickness defined between said first continuous surface and said second continuous surface less than about 0.02 inches, wherein the thickness is less than a width of said first continuous surface and said second continuous surface, the width extending in a direction transverse a longitudinal axis of the body such that the body is a ribbon-shaped body, and wherein said body is configured to curve in a direction perpendicular to said first continuous surface and said second continuous surface, wherein the stiffener has a stiffener thickness greater than the thickness of the body.
12. The insertion apparatus in accordance with claim 11, wherein the body and stiffener are axially movable through the segments of the guide tube.
13. The insertion apparatus in accordance with claim 12, wherein the guide tube has a substantially rigid body and a pliable tip at a guide tube insertion end.
14. A method of inspecting a rotary machine, the rotary machine defining an annular cavity extending along a circumference, said method comprising: positioning a guide tube in a port of the rotary machine to define a path for an insertion apparatus into the annular cavity, the guide tube having a substantially rigid body and a pliable tip at a guide tube insertion end, the insertion apparatus comprising: an insertion end positionable within the annular cavity and configured to



travel through the annular cavity; a steering end opposite said insertion end; a body extending continuously from said insertion end to said steering end and sized to fit within the annular cavity, wherein said body is formed of a flexible material having a first stiffness and curves along the circumference as said insertion end travels through the annular cavity; a stiffener integrally formed and fixed with respect to said body and extending from said steering end to said insertion end, wherein said stiffener has a second stiffness that is greater than the first stiffness and is configured to resist bending of said body, wherein the guide tube is curved to form an at least partially helical shape; wherein the stiffener is fixed adjacent to one edge of the body; wherein the edge of the body corresponding to the location of the stiffener leads an opposite edge of the body that does not correspond to the stiffener during insertion into the annular cavity; wherein the stiffener is configured to resist bending and prevent the body from folding when being translated into the annular cavity; wherein the body and stiffener comprise a uniform and uninterrupted cross-section extending from the insertion end to the steering end; at least one maintenance device coupled to said insertion end of said insertion apparatus; and a displacement mechanism configured to adjust a position of said at least one maintenance device relative to said body; positioning the insertion apparatus along the path and into the annular cavity; guiding the body of the insertion apparatus through the guide tube such that the substantially rigid body of the guide tube does not deform during guiding of the insertion apparatus therethrough and the pliable tip is deformable when contacted by the insertion apparatus; directing the insertion end of the insertion apparatus through the annular cavity without controlling the stiffener; and positioning the at least one maintenance device adjacent a portion of the rotary machine using the displacement mechanism; wherein (1) arrangement of the stiffener fixed and adjacent to one edge of the body and (2) the integral formation of the body with the stiffener are cooperatively effective to provide the body with an overall stiffness, the overall stiffness being constant and unchanged prior to deformation of the body and after deformation of the body as the body curves along the circumference during insertion and movement of the body through the annular cavity.

15. The method in accordance with claim 14, wherein said at least one maintenance device comprises at least one of the following: a mechanical sensor, a thermal sensor, a magnetic sensor, an acoustic sensor, and an electromagnetic sensor.

16. The method in accordance with claim 14, wherein said guide tube includes a plurality of positionable segments.

17. The method in accordance with claim 14, wherein said guide tube is coupled to the port and is sized and shaped to extend from the port to the annular cavity, said guide tube configured to guide said insertion apparatus through the port and into the annular cavity.

18. The method in accordance with claim 14, wherein positioning the at least one maintenance device adjacent a portion of the rotary machine using a displacement mechanism configured to adjust a position of the at least one maintenance device relative to the body comprises pivoting the at least one maintenance device about the insertion end of the insertion apparatus.

19. The method in accordance with claim 14, wherein positioning the at least one maintenance device adjacent a portion of the rotary machine using a displacement mechanism configured to adjust a position of the at least one maintenance device relative to the body comprises inflating a bladder coupled to the at least one maintenance device.

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